

Bike Store Sales Analysis Report

Exploratory Data Analysis (EDA) Report

Project Overview

The objective of this project is to analyze three years of bike store sales data and provide meaningful business insights through SQL-based analysis and an interactive Excel dashboard.

The dataset contains information related to orders, products, brands, categories, stores, customers, and staff. SQL was used to perform data extraction, aggregation, and business analysis. Multiple views were created in SQL and then imported into Excel for visualization.

Data Processing Workflow

Step 1: Data Extraction

Data was collected from multiple relational tables including:

- Orders
 - Order Items
 - Products
 - Categories
 - Brands
 - Stores
 - Staffs
 - Customers
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Step 2: SQL Analysis

SQL queries were written to:

- Calculate Total Sales
- Calculate Total Orders
- Determine Average Order Value (AOV)

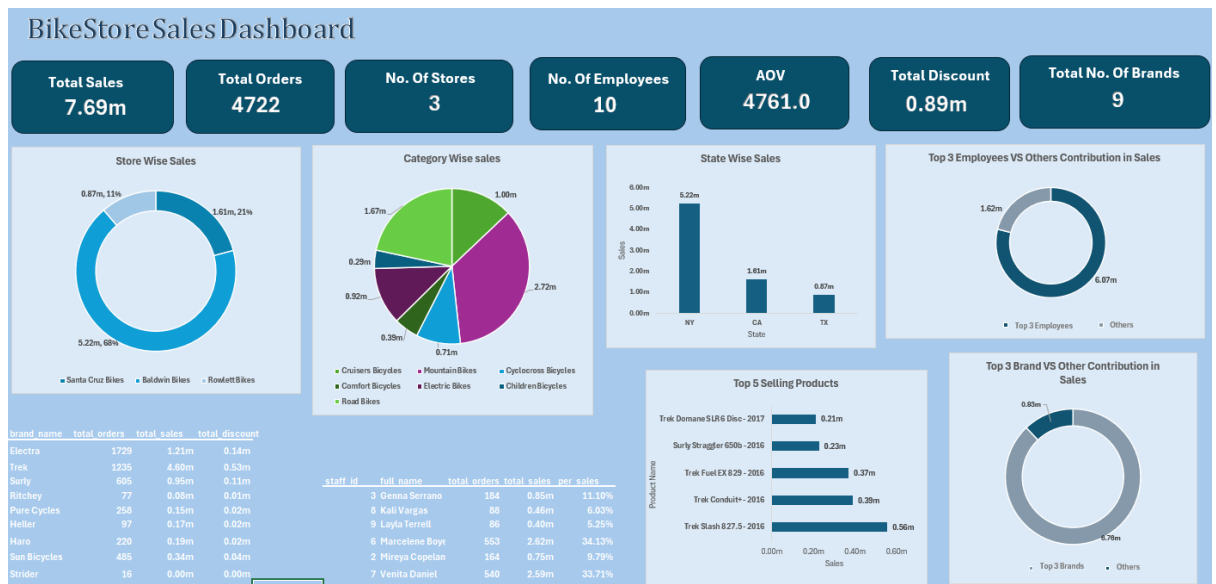
- Analyze Store-wise Performance
- Analyze State-wise Sales
- Analyze Category-wise Sales
- Identify Top Selling Products
- Evaluate Employee Performance
- Compare Brand Contributions
- Calculate Total Discounts

Views were created for each analysis to simplify reporting and dashboard creation.

Step 3: Dashboard Development

The SQL views were imported into Excel and transformed into an interactive dashboard using:

- Doughnut Charts
- Pie Charts
- Bar Charts
- KPI Cards



Key Performance Indicators (KPIs)

Metric	Value
Total Sales	\$7.69M
Total Orders	4,722
Number of Stores	3
Number of Employees	10
Average Order Value	\$4,761
Total Discount	\$0.89M
Number of Brands	9

Sales Analysis

Store-wise Sales Performance

Store	Sales
Baldwin Bikes	\$5.22M
Santa Cruz Bikes	\$1.61M
Rowlett Bikes	\$0.87M

Insights

- Baldwin Bikes contributes approximately 68% of total revenue.
 - Santa Cruz Bikes contributes 21%.
 - Rowlett Bikes contributes 11%.
 - Baldwin Bikes is the highest-performing store.
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Category-wise Sales Analysis

Top Categories

- Electric Bikes → \$2.72M
- Comfort Bicycles → \$1.67M
- Cruisers Bicycles → \$1.00M

Insights

- Electric Bikes generate the highest revenue.
- Premium bike categories contribute significantly to overall sales.
- Children's bicycles generate comparatively lower revenue.

State-wise Sales Analysis

State Sales

NY \$5.22M

CA \$1.61M

TX \$0.87M

Insights

- New York is the dominant market.
- California holds the second position.
- Texas contributes the least sales among the three states.

Product Analysis

Top 5 Selling Products

Product	Sales
Trek Slash 8 27.5 - 2016	\$0.56M
Trek Conduit+ - 2016	\$0.39M
Trek Fuel EX 8 29 - 2016	\$0.37M
Surly Straggler 650b - 2016	\$0.23M
Trek Domane SLR 6 Disc - 2017	\$0.21M

Insights

- Trek products dominate the top-selling product list.
- High-end mountain and electric bikes contribute significantly to sales.

Employee Performance Analysis

Top Performing Employees

Employee	Sales
Mireya Copeland	\$2.59M
Marcelene Boye	\$2.62M
Genna Serrano	\$0.85M

Insights

- Top 3 employees contribute a major share of total revenue.
- Employee performance varies significantly across staff members.
- Sales concentration indicates opportunities for training and performance improvement.

Brand Performance Analysis

Brand Sales

Trek	\$4.60M
Electra	\$1.21M
Surly	\$0.95M

Insights

- Trek is the leading brand by revenue.
- Trek alone contributes nearly 60% of total sales.
- Brand diversification opportunities exist to reduce dependency on a single brand.

Discount Analysis

Findings

- Total Discounts Offered: \$0.89M

- Discounts account for approximately 11.5% of total sales.

Insights

- Discounting plays an important role in driving sales.
 - Further analysis can determine whether discounts positively impact profitability.
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Business Recommendations

1. Expand Trek Inventory

Since Trek products dominate sales, maintaining sufficient stock levels can prevent lost sales opportunities.

2. Improve Low-Performing Stores

Analyze operational differences between Baldwin Bikes and other stores to improve performance.

3. Focus on Electric Bikes

Electric bikes generate the highest category revenue and should receive greater marketing investment.

4. Employee Training Programs

Implement training programs based on the practices of top-performing employees.

5. Review Discount Strategy

Evaluate discount effectiveness to maximize profitability while maintaining sales growth.

Conclusion

The Bike Store Sales Dashboard successfully provides insights into store performance, product demand, employee productivity, brand contribution, and regional sales trends. By leveraging SQL for data analysis and Excel for visualization, the project transforms raw transactional data into actionable business intelligence that supports data-driven decision-making.